

Rehan Sanjay Venkatesan

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Profile

Voice AI and real-time media engineer. **Ten fixes merged** into the frameworks the industry builds voice agents on — eight into **livekit/agents** (two by its co-founder), plus **pipecat** and **jambonz**. One is in livekit's **core connection pool**, fixing a latent bug in eleven of its plugins at once. Built a Twilio-to-Deepgram voice agent end to end, so I know these failures from both sides: the dropped call, and the patch to the framework that dropped it.

Open Source

livekit/agents: **Eight merged** into a 14k-star framework. [#7023](#) an STT connection pool that stayed open after shutdown (merged by David Zhao, co-founder); [#7012](#) per-stream HTTP sessions leaking through a WeakSet, with four regression tests; [#7050](#) a discarded prewarm task rebuilding a pool *after* close; [#7054](#) found the whole bug class, fixed all fifteen instances across library, examples and tests, and got ruff's RUF006 **enabled in their CI** so it cannot return.

core: [#7139 merged into the framework itself](#) — `ConnectionPool.invalidate()` closed sockets that were still streaming, and a handshake in flight escaped invalidation entirely. **Eleven plugins** call it from `update_options`, so one change to the primitive fixed all of them. Reviewed line by line by a maintainer. The same stale-handshake bug, found and fixed in three more provider plugins: [#7132](#) `asyncai`, [#7133](#) `neuphonic`, [#7140 `cartesia`.](#)

pipecat: **Merged** — [#5464](#) an AsyncAI TTS service wrote model, voice and language only into its websocket init message, so a runtime change was stored, warned about, and silently never sent. Found by grepping for a bug `class` across every service, not by reading the tracker.

BerriAI/litellm: [#32758](#) (open) — disabling semantic caching left stale fields on the server. Test verified by mutation.

jambonz: **Merged** — [#1585](#) a TTS `stream_resumed` event sent to a different hook path than the one it resumed.

Also open: Six more under review across **livekit/agents**, **pipecat**, **jambonz**, **drachtio-srf** and **drizzle-orm** — SIP, telephony, real-time media, SQL generation. Each found by reading unfamiliar code, each with a failing test.

Projects

Atlas: AI voice agent platform, ~31,000 lines, built solo. Twilio Media Streams into Deepgram streaming STT over WebSocket, with barge-in, a call-session state machine, outcome classification, outbound dialling inside legal calling hours, and SMS/WhatsApp on one session model. Completed Twilio business verification. NestJS, Next.js 15, Postgres/Prisma, Redis/BullMQ.

InvoiceCheck.in: Live B2B SaaS with payments. 11-point GST compliance check run before marketplaces reject an invoice; OCR auto-fill via OpenCV and Tesseract; Razorpay pay-per-check. Next.js, TypeScript, Supabase.

Coral Reef Inspector: CNN classifying coral bleaching from 5,000+ underwater images combined with pH and temperature readings; a custom OpenCV augmentation pipeline tripled the usable dataset. Python, TensorFlow, Flask.

Experience

Codtech IT Solutions

Backend Web Development Intern

Java backend modules against MySQL — CRUD, authentication, error handling. Designed, documented and tested REST APIs. Built secure login for the Student Management System.

Chennai

Feb–Mar 2025

Makos Infotech

Front End Development Intern

Responsive, cross-browser UI components in React, HTML5 and CSS3, to Agile practice.

Chennai

Jun–Jul 2024

Skills

Voice: Twilio Programmable Voice & Media Streams, SIP, WebRTC, Deepgram streaming STT/TTS, LiveKit Agents, Pipecat, jambonz, drachtio

Languages: Python (asyncio, aiohttp, pytest), TypeScript, JavaScript, Java, SQL

Backend: NestJS, Express, Flask, Prisma, Postgres, MySQL, MongoDB, Supabase, Redis, BullMQ, REST APIs

Frontend: React 19, Next.js 15, Tailwind

Other: Docker, CI/CD, Vercel, Git, TensorFlow, OpenCV, Tesseract OCR, LLM APIs (Anthropic, OpenAI)

Education

Sri Sairam Institute of Technology

B.E. Computer Science and Engineering, Percentage 7.24

Class XII: Sowdambikaa Boys Matric. Hr. Sec. School, 2022 — 87.8%.

Chennai

2022–2026